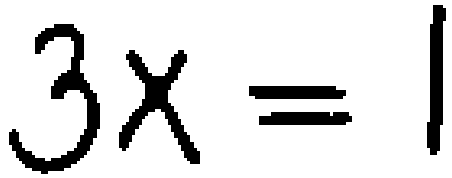
lecture_01_10423__b0__formula

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-S

Render: rendered | Strict EM: True

$$3x = 1$$

```
3x=1
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: True

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-L

Render: rendered | Strict EM: True

$$3x = 1$$

```
3x=1
```

UniMERNet

Render: rendered | Strict EM: True

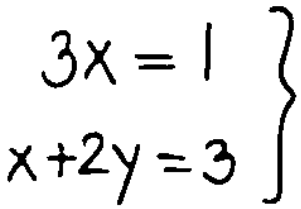
$$3x = 1$$

```
3x=1
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_01_10423__b0__formula_system

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

PP-FormulaNet-S

Render: failed | Strict EM: False

```
3left+x=1\right\
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$3x = 3$$

```
3x=3
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$\{3x = 1\}\{x + 2y = 3\}$$

```
\{3x=1\}\{x+2y=3\}
```

UniMERNet

Render: rendered | Strict EM: False

$$x + 2y = 3$$

```
x+2y=3
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_02_44601__b0__formula

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$x + 3y = 4$$

```
x+3y=4
```

PP-FormulaNet-S

Render: rendered | Strict EM: True

$$x + 3y = 4$$

```
x+3y=4
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: True

$$x + 3y = 4$$

```
x+3y=4
```

PP-FormulaNet-L

Render: rendered | Strict EM: True

$$x + 3y = 4$$

```
x+3y=4
```

UniMERNet

Render: rendered | Strict EM: True

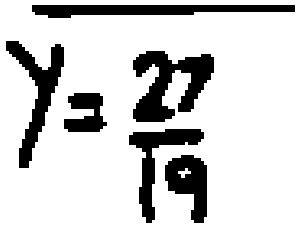
$$x + 3y = 4$$

```
x+3y=4
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_02_44601__b0__formula_fraction

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

PP-FormulaNet-S

Render: rendered | Strict EM: True

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$y = \frac{2pi}{sqrtg}$$

```
y=\frac{2pi}{sqrt{g}}
```

PP-FormulaNet-L

Render: rendered | Strict EM: True

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

UniMERNet

Render: rendered | Strict EM: True

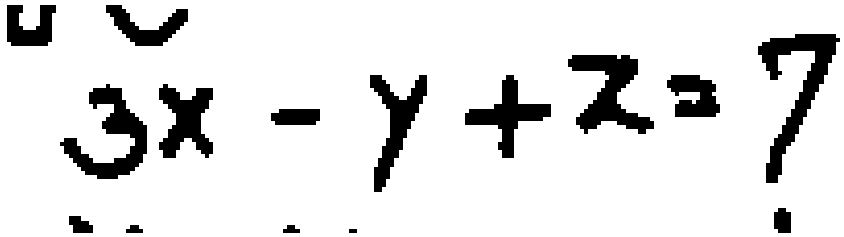
$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

nm_lecture_01_19899__b0__formula

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet-S

Render: rendered | Strict EM: True

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$3x - y + x = 7$$

```
3x-y+x=7
```

PP-FormulaNet-L

Render: rendered | Strict EM: True

$$3x - y + z = 7$$

```
3x-y+z=7
```

UniMERNet

Render: rendered | Strict EM: True

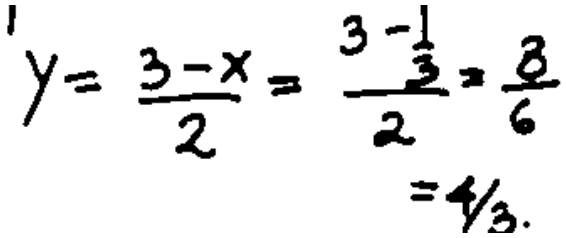
$$3x - y + z = 7$$

```
3x-y+z=7
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

nm_lecture_01_19899__b0__formula_fraction

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

PP-FormulaNet-S

Render: rendered | Strict EM: True

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{3}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{3}{6}
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}=4_{3.}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}=4_{3.}
```

UniMERNet

Render: rendered | Strict EM: True

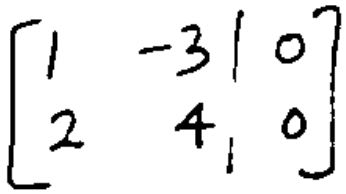
$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_06_47799__b0__augmented_matrix

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

PP-FormulaNet-S

Render: rendered | Strict EM: False

$$[2, 310]]$$

```
[2,310]]
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$[2, \ frac - 3]0o]$$

```
[2,\ frac-3]0o]
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$[1 - 310]$$

```
[1-310]
```

UniMERNet

Render: rendered | Strict EM: False

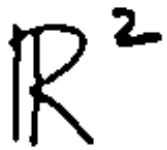
$$\left[\begin{array}{l l l}{1}&{-3\vert}&{0}\\ {2}&{4}&{0}\end{array}\right]$$

```
[\begin{array}{l l l}{1}&{-3\vert}&{0}\\ {2}&{4}&{0}\end{array}]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_06_47799__b0__real_space

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\mathbb{R}^2$$

```
\mathbb{R}^{2}
```

PP-FormulaNet-S

Render: rendered | Strict EM: False

$$R^z$$

```
R^{z}
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$P^z$$

```
P^{z}
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$R^2$$

```
R^{2}
```

UniMERNet

Render: rendered | Strict EM: False

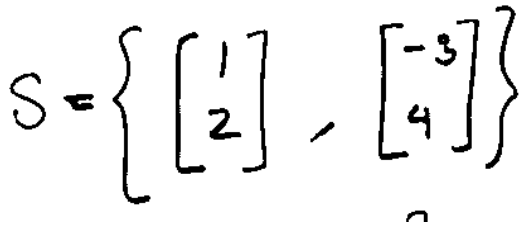
$$R^2$$

```
R^{2}
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_06_47799__b0__vector_set

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$S=\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}-3\\4\end{bmatrix}\right\}$$

```
S=\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}-3\\4\end{bmatrix}\right\}
```

PP-FormulaNet-S

Render: unsupported | Strict EM: False

```
S=\int\[\frac{1}{\sum2}]_{j}^{-1}]_delta{j}^{-}}\\\\ endend
```

PP-FormulaNet_plus-S

Render: unsupported | Strict EM: False

```
s to int limits{{}}}_{{d}}sim{{}
```

PP-FormulaNet-L

Render: unsupported | Strict EM: False

```
S=\{[1\atop2],[4]\}
```

UniMERNet

Render: rendered | Strict EM: False

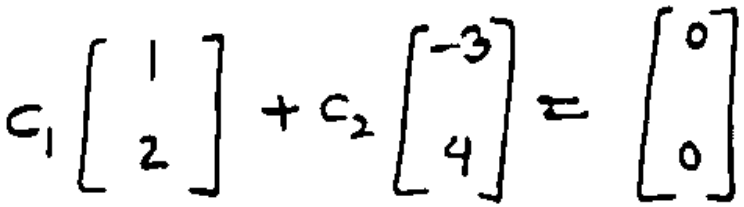
$$S=\{[\frac{1}{2}],[\frac{-3}{4}]\}$$

```
S=\{[\frac{1}{2}],[\frac{-3}{4}]\}
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_06_47799__b0__linear_combination

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

```
c_1\begin{bmatrix}1\\2\end{bmatrix}+c_2\begin{bmatrix}-3\\4\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}
```

PP-FormulaNet-S

Render: rendered | Strict EM: False

$$c_1[\frac{1}{2}] + c_2[\frac{-1}{2}] = \sum 0limits_{0}^{1}$$

```
c_{1}[\frac{1}{2}]+c_{2}[\frac{-1}{2}]=\sum0limits_{0}^{1}
```

PP-FormulaNet_plus-S

Render: unsupported | Strict EM: False

```
a_{1}[i\ frac{1}{z2}]+a_{2}[\frac{3}{z}]^{-3}[\]{2}[\frac{1}{z}]
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$c_1[2] + c_2[4] = [0]$$

```
c_{1}[2]+c_{2}[4]=[0]
```

UniMERNet

Render: rendered | Strict EM: False

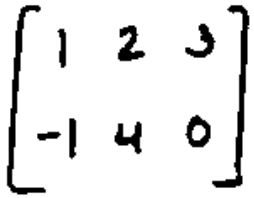
$$c_1[2] + c_2[3] = [0]$$

```
c_{1}[2]+c_{2}[3]=[0]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_07_67415__b0__rectangular_matrix

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\-1&4&0\end{bmatrix}
```

PP-FormulaNet-S

Render: rendered | Strict EM: False

$$[123]$$

```
[123]
```

PP-FormulaNet_plus-S

Render: unsupported | Strict EM: False

```
[int\limits_{[1Object{]}}^{[]2}]^{T}}\\ {\ 1\ 4H}\\ \end{aligned}
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$[123]$$

```
[123]
```

UniMERNet

Render: rendered | Strict EM: False

$$[-140]$$

```
[-140]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_07_67415__b0__symbolic_vector

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

PP-FormulaNet-S

Render: rendered | Strict EM: False

$$\int\limits_{i}^{p}$$

```
\int\limits_{i}^{p}
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$\int\limits_{}^{g} int frac g \frac{g}{z}$$

```
\int\limits_{}^{g}int frac{{g}}{{\frac{g}{z}}}
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$[\frac{p}{q}]$$

```
[\frac{p}{q}]
```

UniMERNet

Render: rendered | Strict EM: False

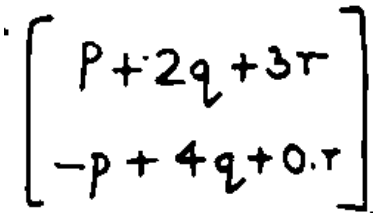
$$[\frac{p}{r}]$$

```
[\frac{p}{r}]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_07_67415__b0__product_vector

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\begin{bmatrix} p+2q+3r \\ -p+4q+0\cdot r \end{bmatrix}$$

```
\begin{bmatrix}p+2q+3r\\-p+4q+0\cdot r\end{bmatrix}
```

PP-FormulaNet-S

Render: unsupported | Strict EM: False

```
\int\limits_{p+2q+3r}^{\ p+4q+0.r}^{\
```

PP-FormulaNet_plus-S

Render: unsupported | Strict EM: False

```
[sqrt r geq2q+3\pi]^{-}]}{(-p+4q+0.r)}_{0}^{-1}
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$[\frac{p+2q+3r}{-p+4q+0.r}]$$

```
[\frac{p+2q+3r}{-p+4q+0.r}]
```

UniMERNet

Render: rendered | Strict EM: False

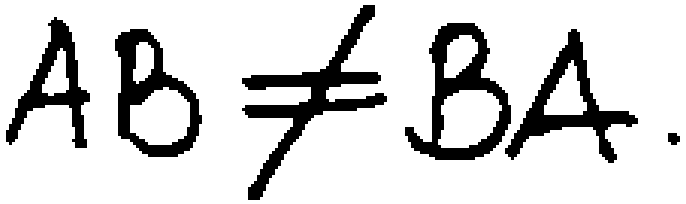
$$[P+2q+3r]$$

```
[P+2q+3r]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_07_67415__b0__noncommuting_product

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$AB \neq BA.$$

```
AB\ne BA.
```

PP-FormulaNet-S

Render: rendered | Strict EM: False

$$AB \neq \beta A.$$

```
A B\neq\beta A.
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$AB \neq BA.$$

```
A B\neq B A.
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$AB \neq BA.$$

```
A B\neq B A.
```

UniMERNet

Render: rendered | Strict EM: False

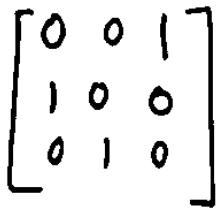
$$AB \neq BA.$$

```
A B\neq B A.
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_08_53016__b0__permutation_matrix

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

PP-FormulaNet-S

Render: unsupported | Strict EM: False

```
\left.\begin{matrix}{0.01}\\ {0.00}\\ {0.0}\\ {0.0}\\ {end0}right\right
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$\begin{matrix}001\\ [1000]\\ 010\end{matrix}$$

```
[\begin{matrix}{001}\\ {1000}\\ {010}\\ \end{matrix}]
```

UniMERNet

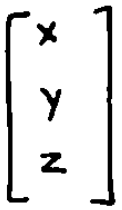
Render: failed | Strict EM: False

```
\left[\begin{array}{l l}{0}&{0}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_08_53016__b0__input_vector

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$\begin{bmatrix}x\\y\\z\end{bmatrix}$$

```
\begin{bmatrix}x\\y\\z\end{bmatrix}
```

PP-FormulaNet-S

Render: unsupported | Strict EM: False

```
\left[\^{x}\right]_{z}
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$\left[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}\right]$$

```
\left[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}\right]
```

UniMERNet

Render: rendered | Strict EM: False

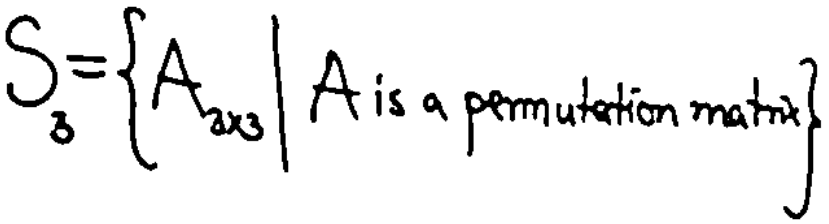
$$\begin{array}{l}{x}\\ {y}\\ {z}\end{array}$$

```
[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}]
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.

lecture_08_53016__b0__set_builder

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$S_3 = \{A_{3\times 3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\left\{A_{3\times3}\mid A\text{ is a permutation matrix}\right\}
```

PP-FormulaNet-S

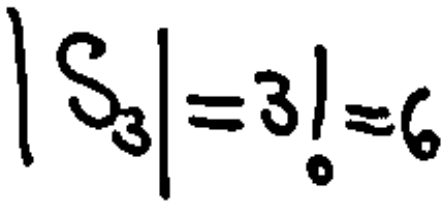
Render: rendered | Strict EM: False

$$S_3 = \{ \lim_{A3 \to A \to B \sin \ !!} \ ! \ !! \ !; !;$$

```
S_{3}=\{\lim\limits_{A3\rightarrow A\rightarrow B\sin\ \ {!!}}\ \ !\ !!\ !;!;\;\;\
```

lecture_08_53016__b0__cardinality_factorial

Input: FCN-LectureNet V34 | Oracle ROI | Full RGB predictions and source LaTeX in comparison.json



Reference

Render: rendered

$$|S_3| = 3! = 6$$

```
|S_3|=3!=6
```

PP-FormulaNet-S

Render: failed | Strict EM: False

```
|vert__{3}\vert=3_{1}=6
```

PP-FormulaNet_plus-S

Render: rendered | Strict EM: False

$$|S_3| = 31 = G$$

```
|S_{3}|=31=G
```

PP-FormulaNet-L

Render: rendered | Strict EM: False

$$|S_3| = 31 = 6$$

```
|S_{3}|=31=6
```

UniMERNet

Render: rendered | Strict EM: False

$$|s_3| = 3| = 6$$

```
\vert s_{3}\vert=3\vert=6
```

Rendering success does not establish correctness. String mismatch may reflect equivalent notation. No GT-based runtime routing.